

Grammatical Evolution: Decoding Integer Array to Expression

Grammar:

$R \mapsto DD'PE$

$D' \mapsto DD' \mid \varepsilon$

$P \mapsto . \, DD' \mid \varepsilon$

$E \mapsto \mathbf{E} S DD' \mid \varepsilon$

$S \mapsto + \mid - \mid \varepsilon$

$D \mapsto 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

Design vector x:

205	52	4	27	10	59	6
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Decoding steps:

R

Only 1 rule $\rightarrow DD'PE$

D (10 options)

$205 \bmod_{10} 10 = 5 \rightarrow \text{digit } \mathbf{5}$

D' (2 options)

$52 \bmod_2 2 = 2 \rightarrow \varepsilon$ (empty)

P (2 options)

$\rightarrow \varepsilon$ (no decimal part)

E (2 options)

$\rightarrow E8$ (exponent)

Result: $4E + 8$

$= 4 \times 10^8$